

WORKING PAPER • PART II OF TWO

A Research Programme on Shared References

Testing and quantifying ad hoc semantic reconciliation across the cognition spectrum



how much does a reference substitute for cognition?

Christopher Janz

A reference partially substitutes for cognition

Part I fixed the terms: a semantic model is knowledge a machine can comprehend; the lift completes each side; reconciliation makes two correspond, at the semantic level. The cognition spectrum — two, one, or none of the systems interrogable — is the axis along which the work, the confidence, and the residual all change.

H1

A reference's marginal benefit grows as machine cognition recedes from the systems being reconciled.

A reference reduces the work of reconciliation in every case — but where it matters most is where reasoning is scarce, standing in for the confirmation a silent system cannot give.

One central claim, five supports

H1 A reference partially substitutes for cognition: benefit grows as cognition recedes.

H2 A thin lexical reference reduces work and surviving false cognates, in every placement.

H3 Different reference types help different operational cases differentially.

H4 Reference benefit shows diminishing returns with richness — thin captures most of it.

H5 References improve confidence calibration and shrink the residual and ratification burden.

H6 Verification available to automation depends on placement, and its effort rises as cognition recedes.

Two tracks, two standards of evidence

ANALYTIC TRACK

It predicts

- An explicit cost model over a case's structural parameters.
- Computes mechanical work, scaling, residual, predicted reliability and confidence.
- Cheap, deterministic; fixes the shape of the metrics.
- Counts operations — it does not measure the cognitive effort they demand.

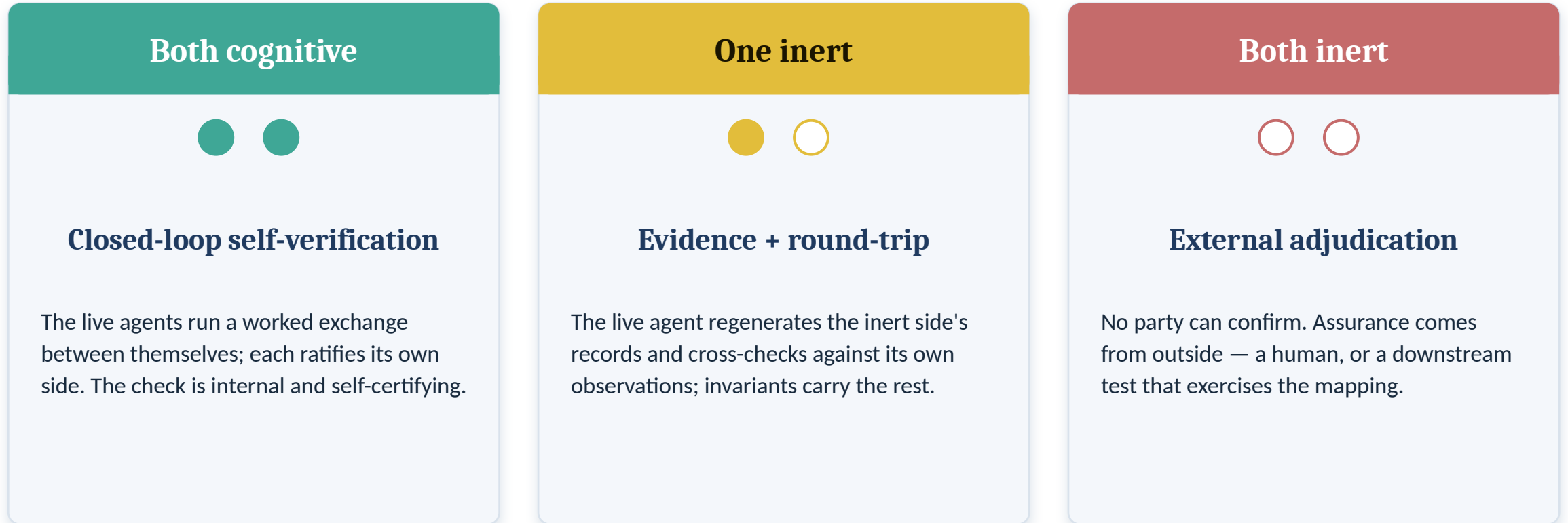
EMPIRICAL TRACK

It measures

- Reasoning systems run on real and synthetic pairs, with and without each reference, at each placement.
- Observes correspondences, surviving false cognates, precision and recall, confidence calibration.
- Records the verification method and the cognitive effort actually expended.
- Where the hypotheses are genuinely tested; reports distributions over repeated trials.

The analytic track predicts what the empirical track then puts to the test.

Which check is available shifts with the cognition



The stronger, closed-loop checks live at the cognitive end. As cognition recedes, the effort of verification rises — measured with the reasoning systems in the loop.

Counting is not measuring

Counting operations captures the mechanical work well. It does not capture the effort a reasoning system expends — especially once language models are in the workflow, where the same nominal step may be trivial or may demand extended deliberation. Effort must be observed, not only counted.

MECHANICAL WORK

Counted

Turns · messages · candidates · checks · operations · scaling
with the number of systems

COGNITIVE EFFORT

Measured, with LLMs in the loop

Tokens & reasoning tokens · tool calls · retries & self-
corrections · latency · a graded difficulty rubric

Effectiveness is paired with effort throughout: not only whether a result is correct, but how much cognition it took to get there.

What we measure, and how the grid is cut

FIVE METRIC FAMILIES

- **Mechanical work** — countable operations
- **Cognitive effort** — measured, LLMs in the loop
- **Scaling** — cost vs. number of systems
- **Reliability** — false cognates, precision/recall, verification
- **Uncertainty & closure** — confidence, calibration, residual, ratification

FOUR FACTORS VARIED

- Reference** — none → thin → +1 type → richer
- Cognition placement** — both / one inert / both inert
- Operational case** — config · intent · x-domain · observ.
- Difficulty & scale** — divergence, cognates, opaque, N

Headline figure
Reference benefit vs. placement, one line per type — the direct test of H1 and H3.

One factor varied at a time from a baseline cell per case; targeted interaction grids fill in per hypothesis.

Used, generated, and shared

Used / consumed

- Published standards & models: TAPI, IETF TE topology, L1CSM & TE service-mapping, TM Forum intent, alarm model, NMOP
- Operator-supplied models and data — external validity

Generated

- Demonstration suite (exists)
- Analytic cost-model instrument (exists)
- Evaluation harness (to build)
- Reasoning stack under test (to build)
- Reference-reconciliation benchmark + gold standards
- Reference-type artifact set
- Result datasets

Shared

- Demonstration suite & analytic instrument
- Metric definitions & cost model
- Benchmark & result datasets — candidate for open release

Four seed pairs, then synthetic families around each



Configuration

Transport API topology ↔ IETF TE topology



Intent

TM Forum intent model ↔ IETF L1CSM



Cross-domain

bespoke transport ↔ bespoke IP-service



Observability

incumbent alarm model ↔ NMOP anomaly + incident

Every pair carries a gold-standard record, in a fixed schema

correct correspondences · known false cognates · residual expected by placement · invariants a correct translation preserves · the verification method the case admits

A generator turns divergence, false cognates, and opaque items as parameters — ground truth known by construction.

Each phase delivers something that stands on its own

0-1

Metrics & analytic experiment

Fixed metric definitions and cost model; the analytic instrument. Largely in hand.

2

Benchmark

Corpus, gold standards (incl. verification method & residual by placement), synthetic generator, reference-type artifacts.

3

Empirical runs

Evaluation harness and reasoning stack; measured runs, capturing cognitive effort and verification as well as work and reliability.

4

Analysis

Effect sizes and interaction plots; the headline figure; measured results compared against the analytic predictions.

Where an operator's contribution matters most

The synthetic benchmark gives clean effect sizes; the real part gives external validity. In rough order of value:

- **Real, bespoke model pairs** — especially with opaque or under-documented fields — the material the inert cases turn on
- **Alarm, event & telemetry streams** — with enough context to reconstruct meaning; the observability case depends on it
- **Intent & service definitions** — as operators actually express them, with real quantitative content
- **Ground-truth reconciliations** — an existing mapping or an adjudication supplies the gold standard the real part lacks
- **A setting to run against operator data** — so measured results reflect real models, not only public ones

One early experiment — reconcile a pair of an operator's own models at each placement, with and without a thin reference — speaks directly to H1, H2, H5, and H6.

SCOPE

A deliberately bounded programme

The programme studies shared references and the cognition spectrum for semantic reconciliation. It does not attempt a universal model, nor a general treatment of agent governance, security, or authorisation — real problems, left to companion work.

Close to current standards activity

The observability case builds directly on recent fault-management and anomaly work. The effect of shared references on human-supervision requirements is intended to be useful to that activity.

Part I · the theory and the terms Part II · the programme to test and quantify them